RICE LEAF SCALD: PATHOGEN BIOLOGY AND DIVERSITY

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1. Introduction

Rice leaf scald is caused by the fungal pathogen *Monographella albescens* (Thüm) Parkinson, Sivanesan and Booth (anamorph *Gerlachia oryzae* (Hashioka and Yokogi) W. Gams). It is a disease that can be found throughout virtually all of the rice growing regions of the world (International Mycological Institute, 1996). Under favourable climatic and cultural conditions, *M. albescens* can pose a serious threat to rice production and is often ranked in the first three or four fungal constraints for a region, along with blast (*Magnaporthe grisea*), sheath blight (*Rhizoctonia* spp.) and brown spot (*Helminthosporium oryzae*) (Jones *et al.*, 1991; Alam *et al.*, 1985; Cuevas-Perez and Gaona, 1988; Kempf, 1983; Prabhu and Sitarama-Prabhu, 1989). Incidence of infection in an area can reach high levels under favourable conditions, resulting in significant loss of yield, such as the 23.4% yield reduction reported in India by Srinivasan (1981).

Much of the literature relating to this disease comprises first reports of the pathogen's presence in a given region, or revisions of the taxonomy of the anamorph and teleomorph. Variations in symptomology and host tissues affected have given rise to a variety of common names for the disease, for example leaf scald, brown leaf blight, leaf tip blight and leaf tip drying (Ou, 1985). A succinct overview of the taxonomy of the pathogen was produced by Ou (1985), while Parkinson *et al.* (1981) have provided good evidence for the current classification, with *Monographella albescens* superseding *Metasphaeria albescens* (Thum) for the teleomorph, and *Gerlachia oryzae* superseding *Rhynchosporium oryzae* (Hashioka and Yokogi) for the anamorph.

Despite the wide geographical distribution of *M. albescens*, this pathogen has been the subject of remarkably little research, being overshadowed by the more damaging rice blast pathogen *Magnaporthe grisea* (anamorph *Pyricularia oryzae*). However, as blast-resistant rice varieties have been developed and introduced into general use around the world, it is worth bearing in mind that leaf scald is a near-ubiquitous disease that can become a severe problem when favourable conditions are prevalent. In light of this, some attention should be devoted to the biology of the causal agent and particularly to

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the levels of genetic and phenotypic variability within and between populations. Current understanding of these aspects is therefore discussed below.

2. Morphology and Culture

M. albescens is a uninucleate, filamentous fungus (Hyponectriaceae, Ascomycota) which is most often observed causing characteristic light-and-dark banded brown lesions which extend down the leaf from the tip or inwards from leaf margins, drying the leaf tissue as the lesion develops (Figure 1, arrowhead). Scald infection can occur in conjunction with other diseases, such as blast (Figure 1, arrow). *M. albescens* has also been shown to infect the leaf sheath, seeds, spikelets and panicles (Ribeiro, 1983).

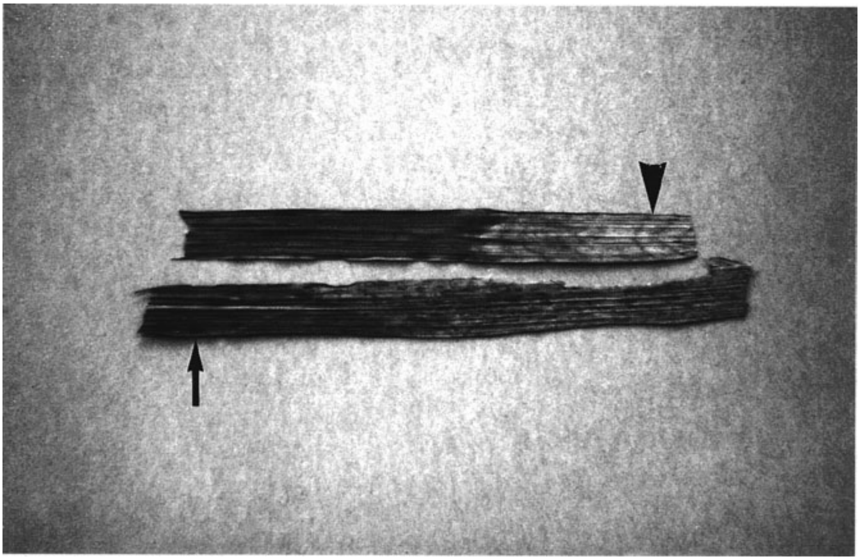

Figure 1. Rice leaves showing typical leaf scald symptoms. Note how the lesion, spreading down from the leaf tip, shows a distinctive banding pattern (arrowhead). These leaves also show signs of infection by the rice blast pathogen *Magnaporthe grisea* (arrow).

The pathogen forms falcate, predominantly bitunicate conidia on the leaf surface (Figure 2); the conidiogenous structures often emerge from stomata. The conidia are 9-14 µm long by 3-4.5 µm wide (Ou *et al.*, 1978). Perithecia, containing colourless paraphyses and sets of eight ascospores per ascus, develop within the mesophyll layer of infected leaves.

M. albescens can be cultured on potato dextrose agar (PDA) where it produces abundant pale pink or cream mycelium (Parkinson, 1980). Conidia form on ageing